

From Data to Solutions



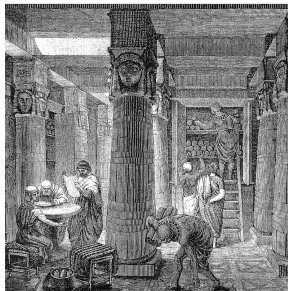
RAG foundations

Weekend 5 · Retrieval Augmented Generation

Friday 14 August 2026

Carlos Cotrini

Alexandria, about 250 BCE



- A scroll has no title page. To learn what it is, you unroll it.

Imagined by O. Von Corven, 19th century.

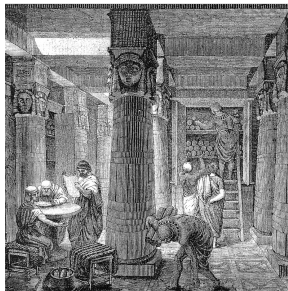
About 250 BCE

121 CE

1230

1895

Alexandria, about 250 BCE



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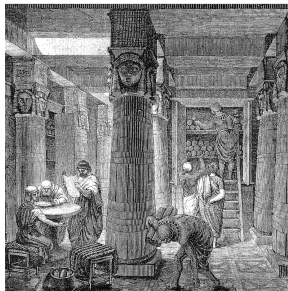
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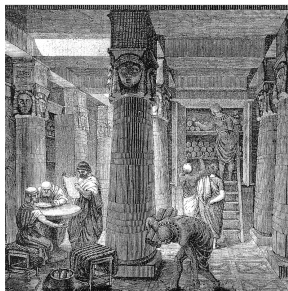
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Alexandria, about 250 BCE

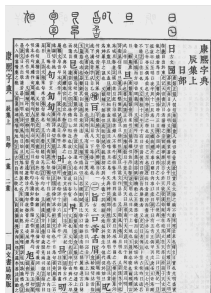


Imagined by O. Von Corven, 19th century.

- A scroll has no title page. To learn what it is, you unroll it.
- **A scholar arrives wanting the works of one author.**
- Callimachus wrote **one entry per work: author, opening words, length in lines**, in 120 scrolls.
- You read the entries until you know which of **the scrolls** to unroll.

China, 121 CE

- A writing system with no alphabet has no order to sort by.



A page of the Kangxi Dictionary, 1716.

About 250 BCE

121 CE

1230

1895

China, 121 CE



- A writing system with no alphabet has no order to sort by.
- **You meet a character you cannot read.**

A page of the Kangxi Dictionary, 1716.

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From Data to Solutions

China, 121 CE



- A writing system with no alphabet has no order to sort by.
- You meet a character you cannot read.
- Xu Shen gave each one an address: **one of 540 components**, and after 1615 **a stroke count too**.

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China, 121 CE



- A writing system with no alphabet has no order to sort by.
- You meet a character you cannot read.
- Xu Shen gave each one an address: **one of 540 components**, and after 1615 **a stroke count too**.
- Follow the address and you reach **the entry: what it means, how it is said**.

A page of the Kangxi Dictionary, 1716.

About 250 BCE

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Paris, 1230



- About 500 friars listed every word of the Bible, and everywhere it appears.

Hugh of Saint-Cher. Treviso, 1352.



Paris, 1230



- About 500 friars listed every word of the Bible, and everywhere it appears.
- **You want every passage that uses one word.**

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- About 500 friars listed every word of the Bible, and everywhere it appears.
- **You want every passage that uses one word.**
- There was no way to say where, so they cut every chapter into **seven parts, marked A to G.**



Paris, 1230



Hugh of Saint-Cher. Treviso, 1352.

- About 500 friars listed every word of the Bible, and everywhere it appears.
- **You want every passage that uses one word.**
- There was no way to say where, so they cut every chapter into **seven parts, marked A to G.**
- The address sends you to **the passages themselves.** We still cite that way.



Brussels, 1895



The Repertoire, about 1900.

- Otlet and La Fontaine began **one index card for every book and article ever printed**: 400,000 in the first year, later over 15 million.



Brussels, 1895



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- **From 1896 you could post them a question, for a fee.**

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The Repertoire, about 1900.

- Otlet and La Fontaine began **one index card for every book and article ever printed**: 400,000 in the first year, later over 15 million.
- **From 1896 you could post them a question, for a fee.**
- They searched the cards and mailed back the ones that matched.

Brussels, 1895



The Repertoire, about 1900.

- Otlet and La Fontaine began **one index card for every book and article ever printed**: 400,000 in the first year, later over 15 million.
- **From 1896 you could post them a question, for a fee.**
- They searched the cards and mailed back the ones that matched.
- The cards told you where to find **the books and the articles**. Over 1,500 questions a year by 1912.

The same three things, four times

Query

Keys

Values

The same three things, four times

	Query	Keys	Values
Alexandria, 250 BCE	The works of one author	Author, opening words, length in lines	The scrolls

The same three things, four times

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A company, 2026	A question in plain words	?	The company's own documents

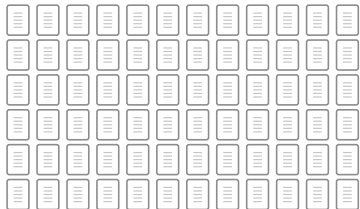
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This hour builds the missing column, so that a language model can answer from those documents.

How do we do retrieval with LLMs?

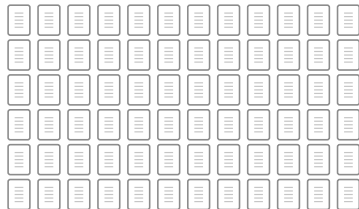
“What did we agree
with the rope supplier
about late delivery?”



all of our documentation

How do we do retrieval with LLMs?

“What did we agree with the rope supplier about late delivery?”



all of our documentation

all of it, at once

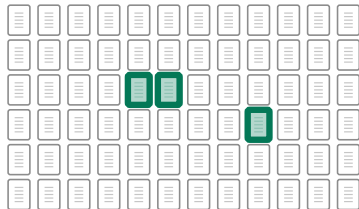


it does not ☐t

LLM

How do we do retrieval with LLMs?

“What did we agree with the rope supplier about late delivery?”



all of our documentation

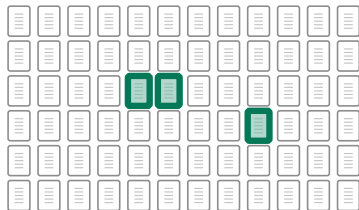


only the few that answer it

LLM

How do we do retrieval with LLMs?

“What did we agree with the rope supplier about late delivery?”



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only the few
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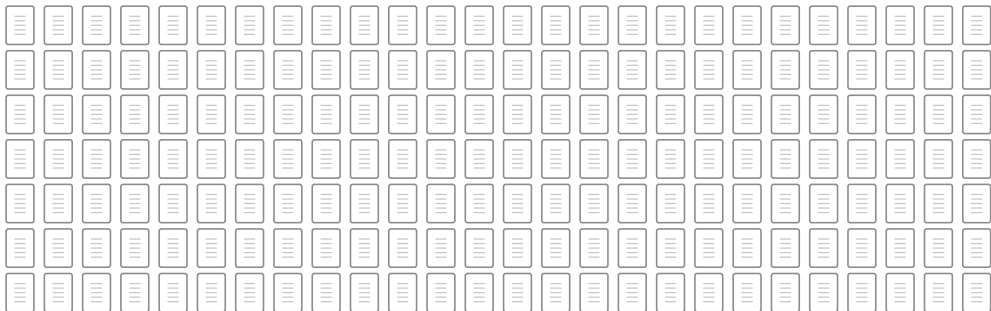
LLM

an answer, from the
documents it was shown

Finding those few, reliably, is what this hour is about.

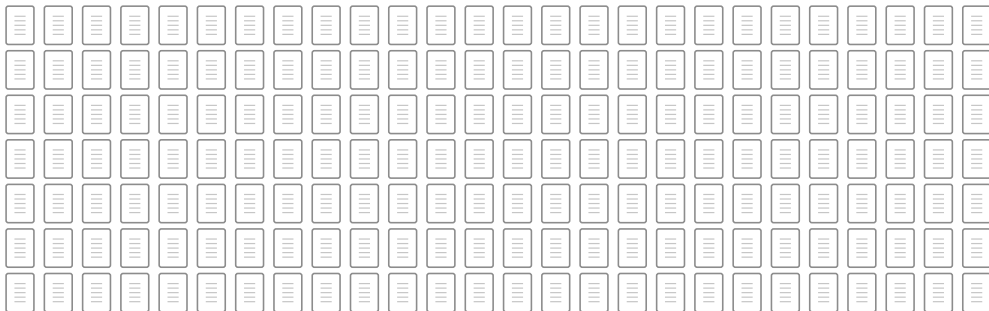
One company's own documents

Finding the few documents that answer a question



One company's own documents

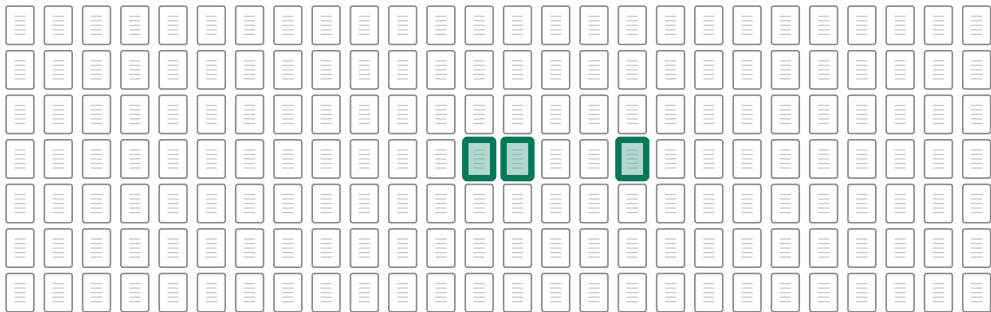
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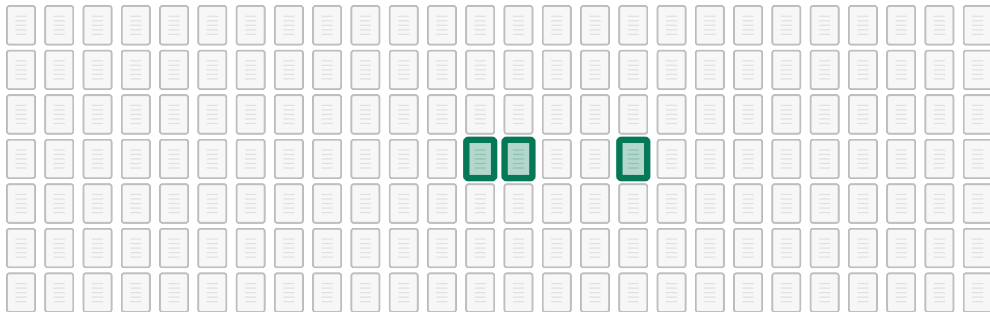
Finding the few documents that answer a question



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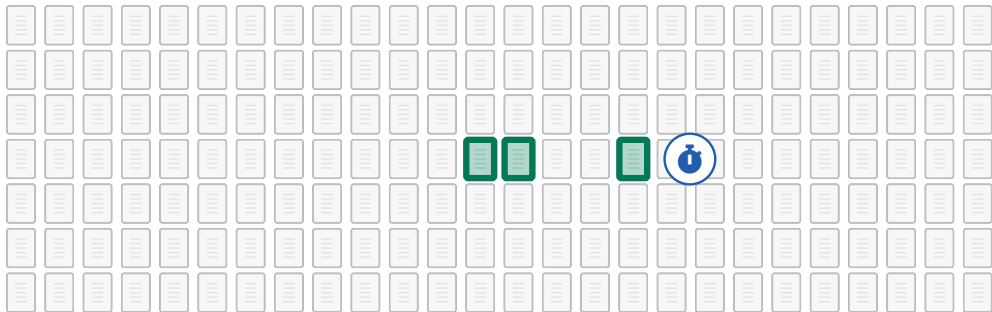
Finding the few documents that answer a question



“What did we agree with the rope supplier about late delivery?”

One company's own documents

Finding the few documents that answer a question



“What did we agree with the rope supplier about late delivery?”

The pub quiz game

1 Restaurant notice	2 Maintenance log	3 Weather and closure rule	4 Service notice
5 Training record	6 Supplier contract clause	7 Weather and closure rule	8 Technical data sheet
9 Lost property record	10 Service notice	11 Weather and closure rule	12 Maintenance log
13 Training record	14 Service notice	15 Sta handbook section	16 Lost property record
17 Supplier contract clause	18 Restaurant notice	19 Lost property record	20 Incident report

twenty of the fifty shown, in no order

The pub quiz game

Vorlen Bergbahnen AG. Fifty internal documents. About 2,600 words.
(a company that does not exist)

1 Restaurant notice

2 Maintenance log

3 Weather and closure rule

4 Service notice

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Fifty documents Six questions, at your own pace

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Six questions, at your own pace

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eth-fdd-fs26.github.io/FDD-Page/document-pile



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The six times you measured

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Restaurant notice

Maintenance log

Weather and closure rule

Service notice

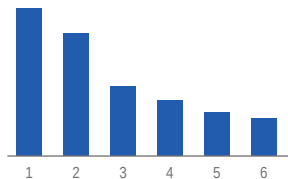
Supplier contract clause

Technical data sheet

Motivation

The six times you measured

your six rounds, on your own screen



illustrative: everyone's diifers

eth-fdd-fs26.github.io/FDD-Page/document-pile

Restaurant notice

Maintenance log

Weather and closure rule

Service notice

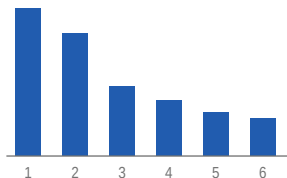
Supplier contract clause

Technical data sheet

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illustrative: everyone's differences

___ of ___ hands: my last round beat my first

Restaurant notice

Maintenance log

Weather and closure rule

Service notice

Supplier contract clause

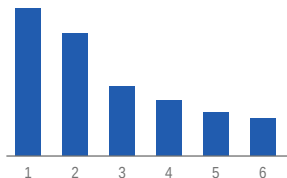
Technical data sheet

eth-fdd-fs26.github.io/FDD-Page/document-pile

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Maintenance log

Weather and closure rule

Service notice

Supplier contract clause

Technical data sheet



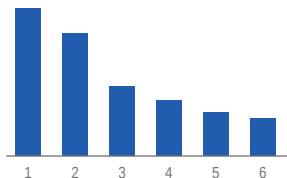
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Maintenance log

Weather and closure rule

Service notice

Supplier contract clause

Technical data sheet



Incident report



Technical data sheet



Maintenance log



Weather and closure rule



Station handbook section



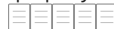
Supplier contract clause



Restaurant notice



Lost property record



Training record



Service notice



Queries, keys and values

VALUES

cards, not chapters



Queries, keys and values

QUERIES

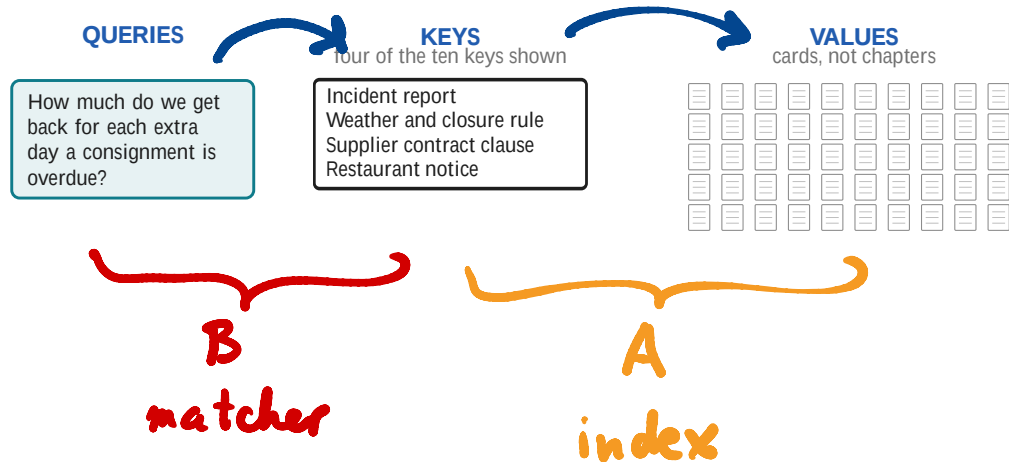
How much do we get back for each extra day a consignment is overdue?

VALUES

cards, not chapters



Queries, keys and values



Queries, keys and values

QUERIES

How much do we get back for each extra day a consignment is overdue?

KEYS

four of the ten keys shown

Incident report
Weather and closure rule
Supplier contract clause
Restaurant notice

VALUES

cards, not chapters



Queries, keys and values

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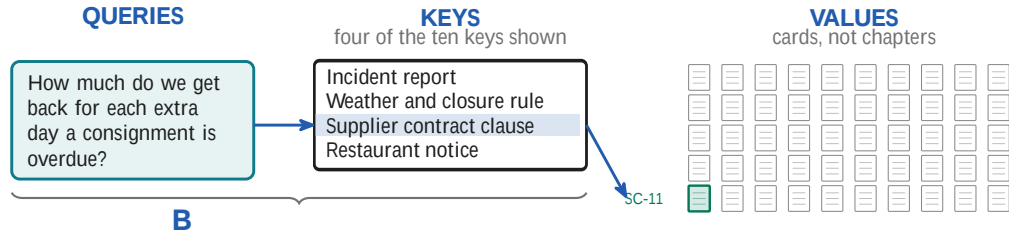
Incident report
Weather and closure rule
Supplier contract clause
Restaurant notice

VALUES

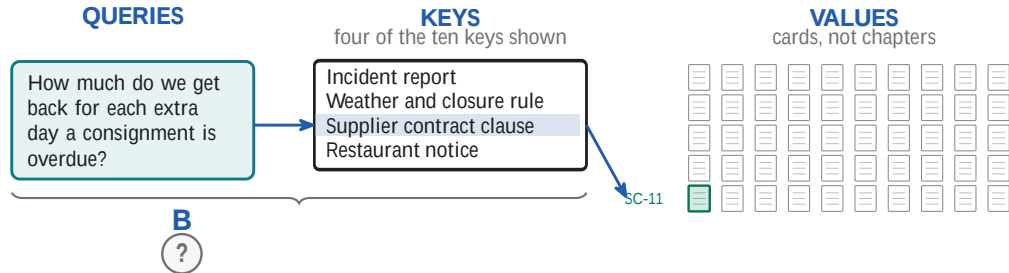
cards, not chapters



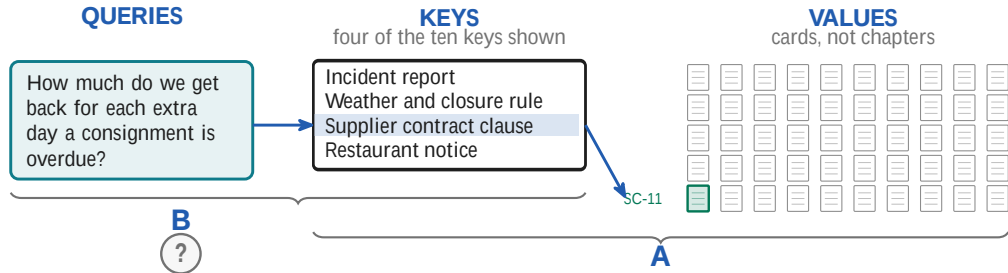
Queries, keys and values



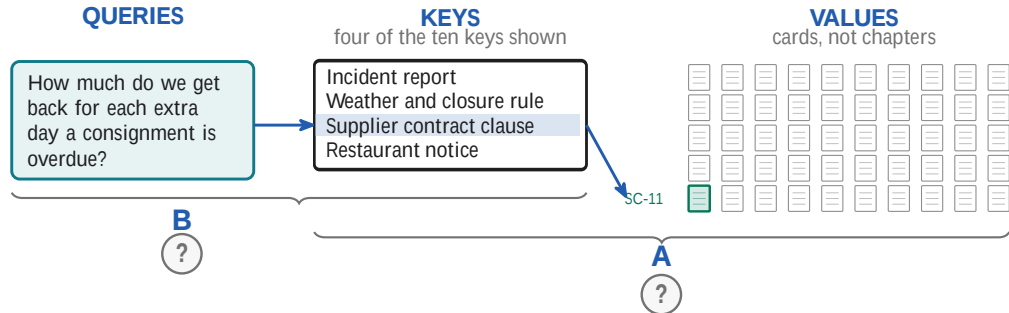
Queries, keys and values



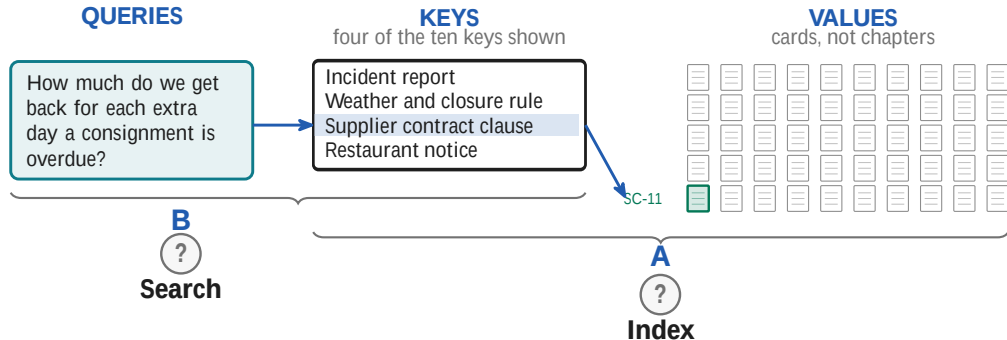
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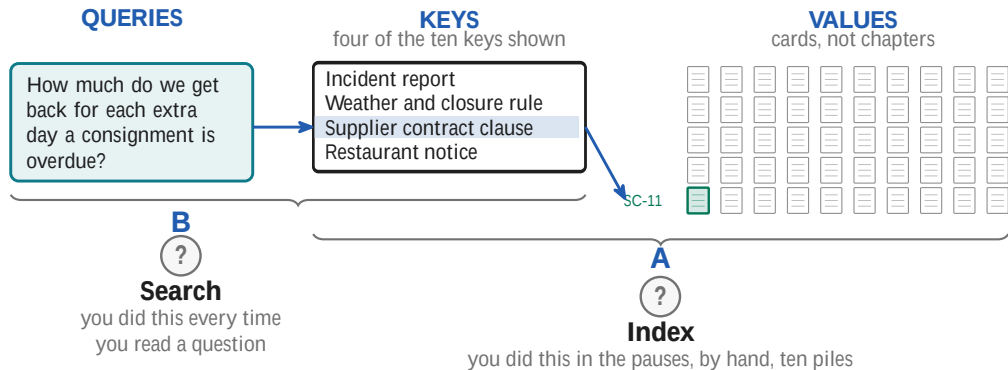
Queries, keys and values



Queries, keys and values



Queries, keys and values



The yogurt cup game



[FDD-Page/yogurt-cups](#)

The yogurt cup game



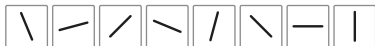
[FDD-Page/yogurt-cups](#)



sender



receiver



The yogurt cup game

- Both see every picture



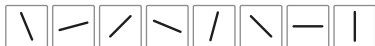
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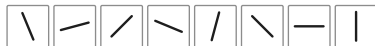
sender



receiver



the same eight
of the twenty four



The yogurt cup game

- Both see every picture
- One sees the target



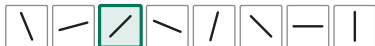
[FDD-Page/yogurt-cups](https://fdd-page.github.io/yogurt-cups)



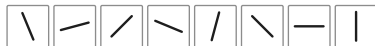
sender



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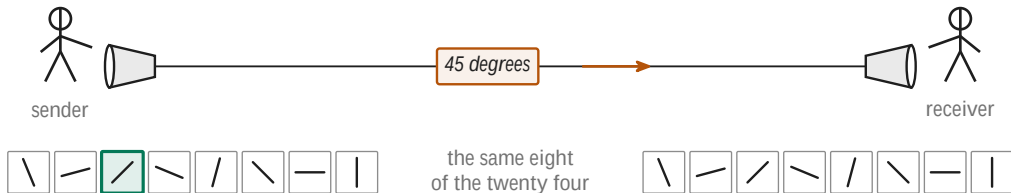


The yogurt cup game

- Both see every picture
- One sees the target
- Five words through the cup



[FDD-Page/yogurt-cups](https://fdd-page.github.io/yogurt-cups)

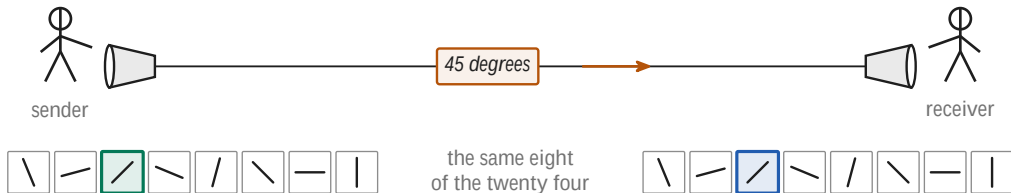


The yogurt cup game

- Both see every picture
- One sees the target
- Five words through the cup
- The partner picks one



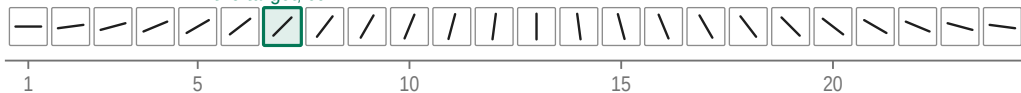
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Partial credit

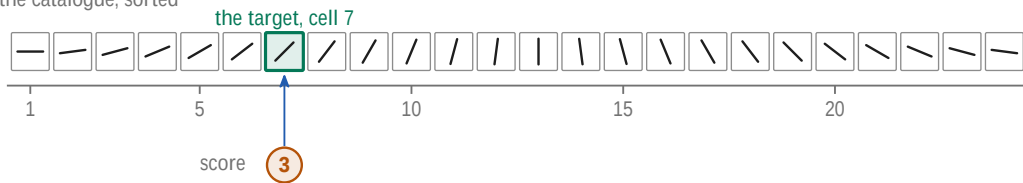
the catalogue, sorted

the target, cell 7



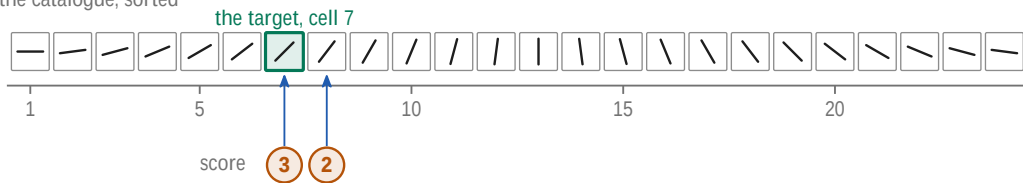
Partial credit

the catalogue, sorted



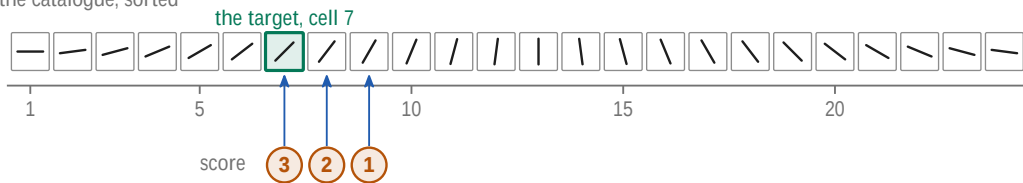
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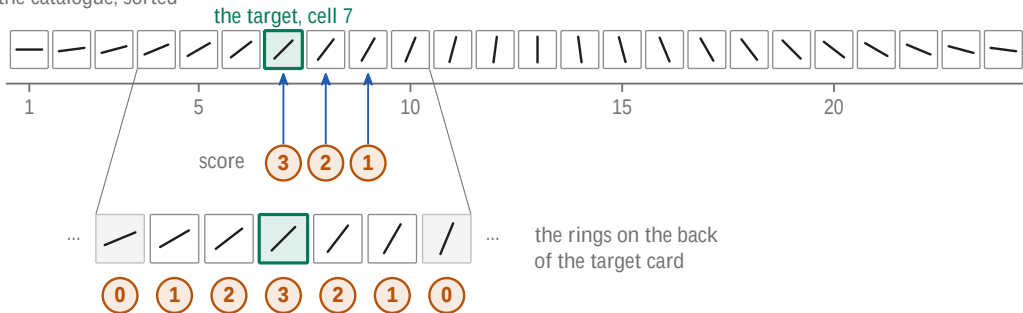
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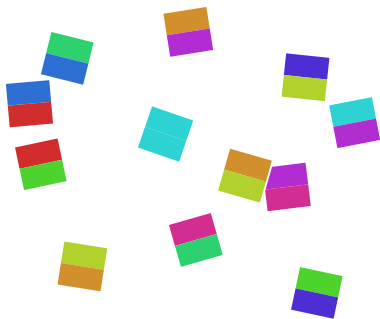


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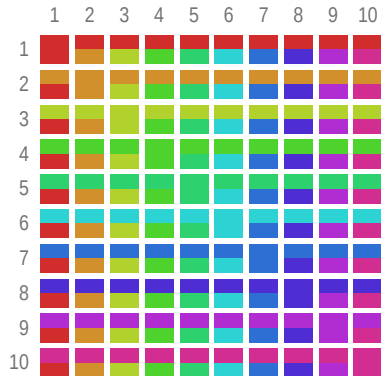


Words are a coarse code

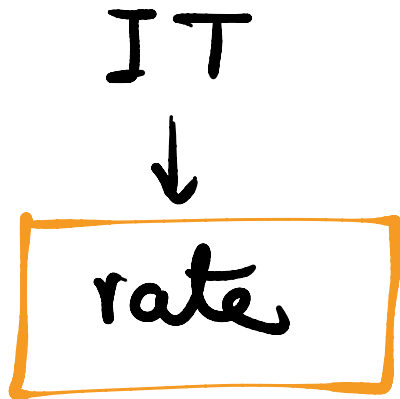


twelve of them, in no order

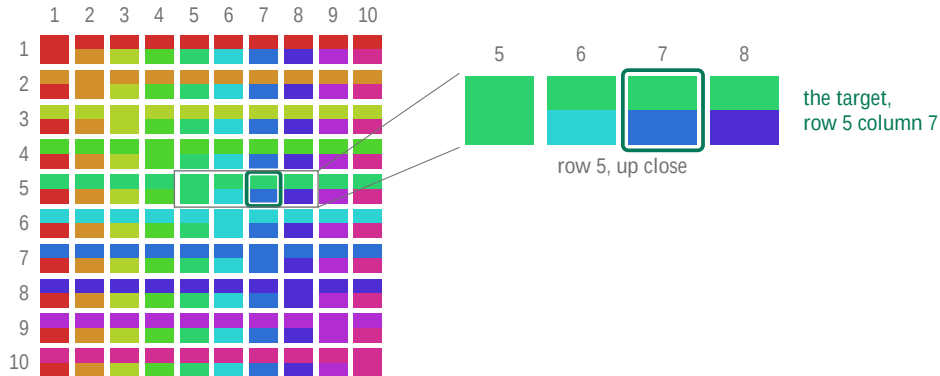
Words are a coarse code



the same ags, sorted on two axes: one hundred, and every one addressable

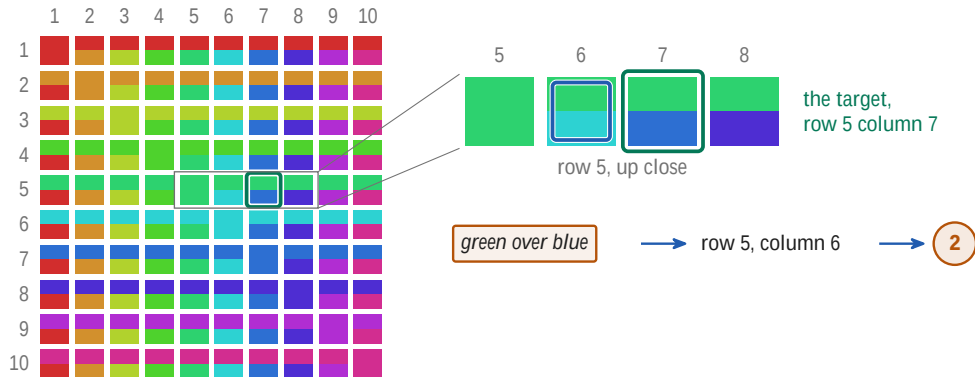


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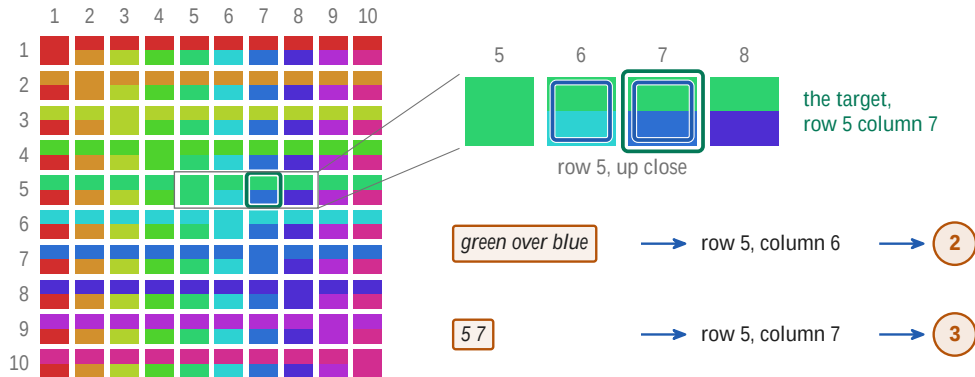
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Words are a coarse code



the same ☐ags, sorted on two axes: one hundred, and every one addressable

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the same ☐ags, sorted on two axes: one hundred, and every one addressable

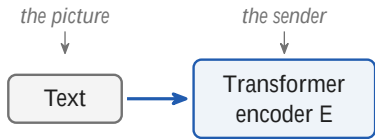
Autoencoders

the picture

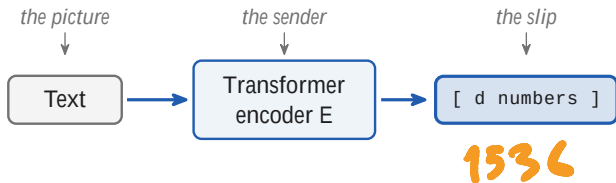


Text

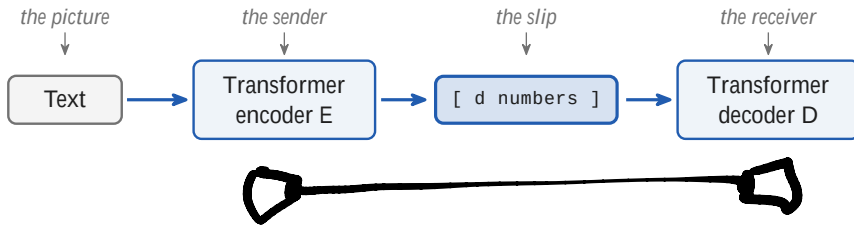
Autoencoders



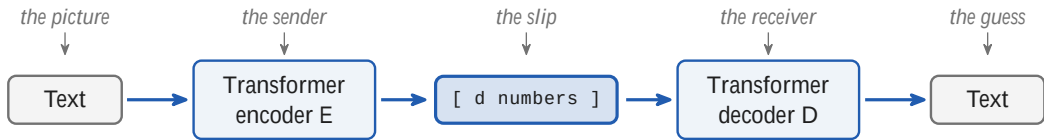
Autoencoders



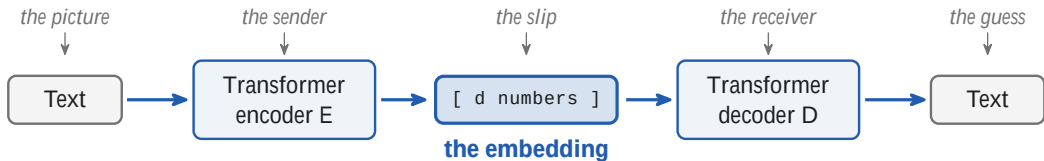
Autoencoders



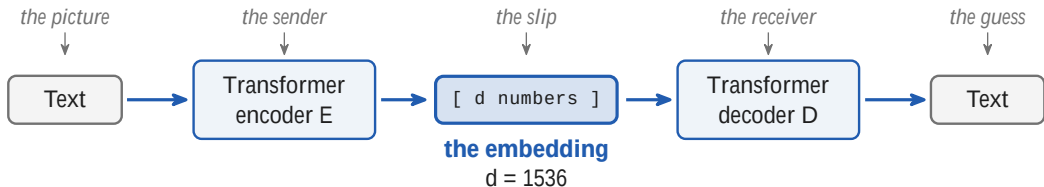
Autoencoders



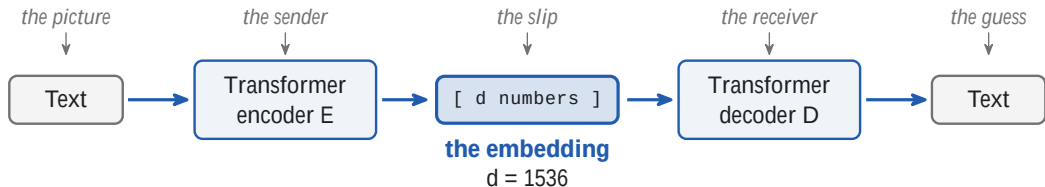
Autoencoders



Autoencoders



Autoencoders



card SC-11

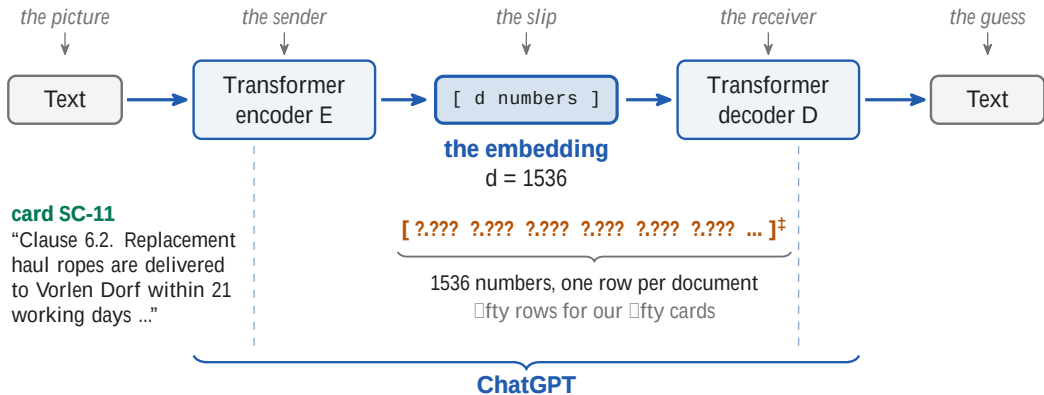
“Clause 6.2. Replacement haul ropes are delivered to Vorlen Dorf within 21 working days ...”

[???? ???? ???? ???? ???? ???? ...][‡]

1536 numbers, one row per document

100 rows for our 100 cards

Autoencoders



How it is trained

- 1 Collect a large collection of texts

How it is trained

- 1 Collect a large collection of texts
- 2 Initialise the weights of E and D at random

How it is trained

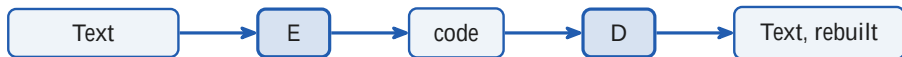
- 1 Collect a large collection of texts
- 2 Initialise the weights of E and D at random
- 3 Play the game, take the score

How it is trained

- 1 Collect a large collection of texts
- 2 Initialise the weights of E and D at random
- 3 Play the game, take the score
- 4 Update the weights from the score, play again

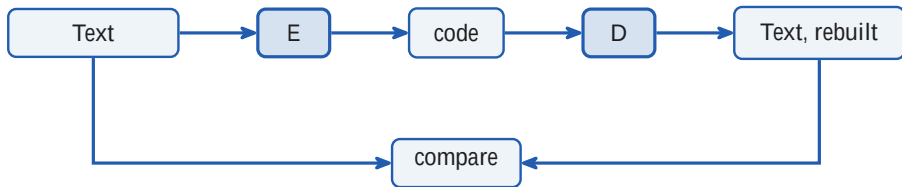
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How it is trained

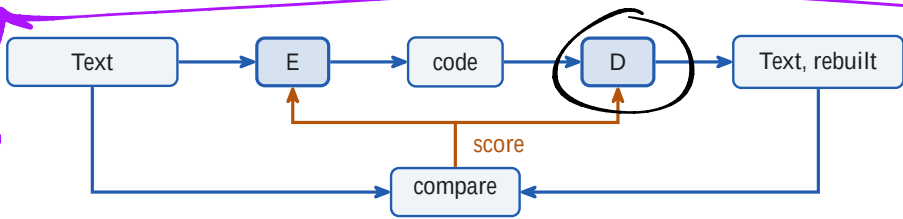
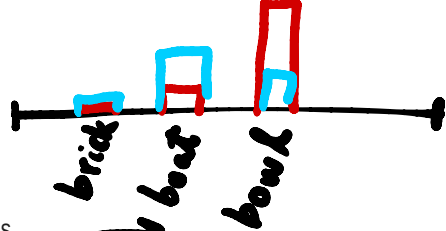
- 1 Collect a large collection of texts
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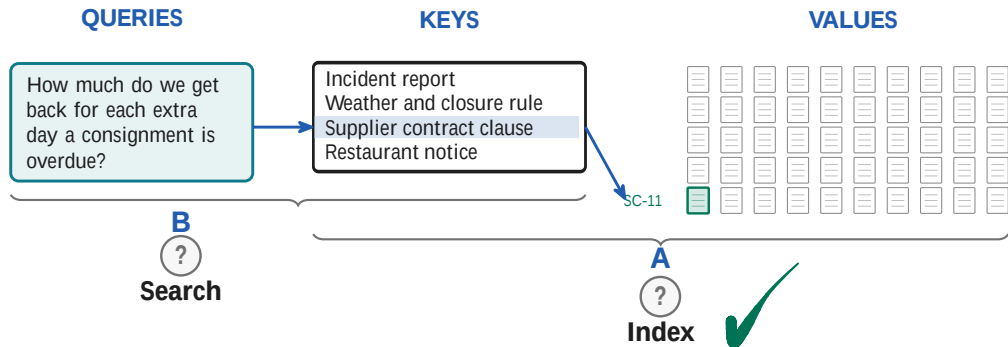
How it is trained

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- 3 Play the game, take the score
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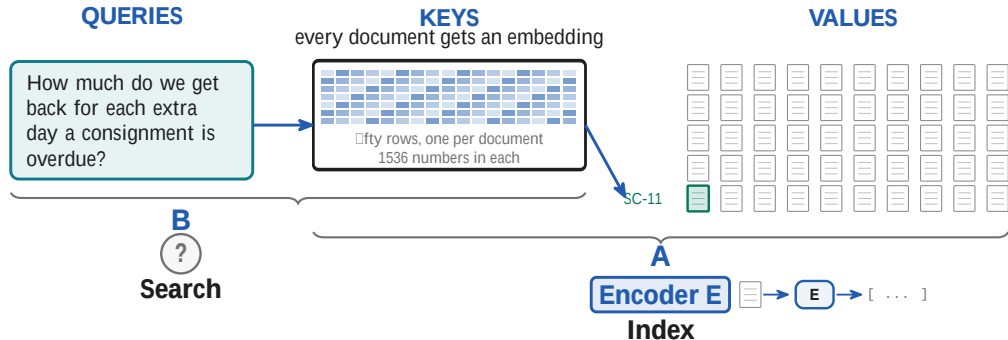
beat



Queries, keys and values

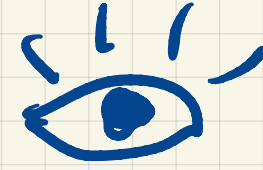
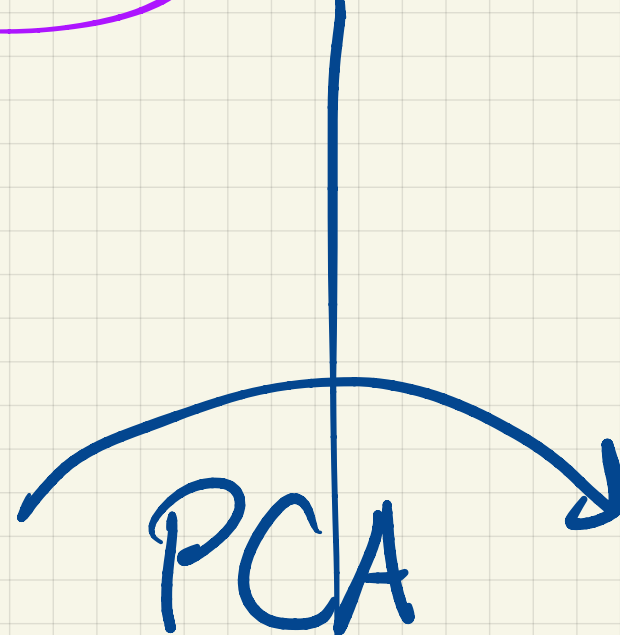
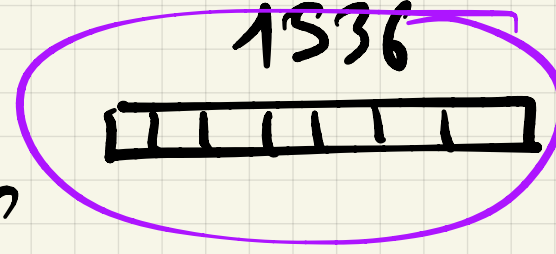


Queries, keys and values



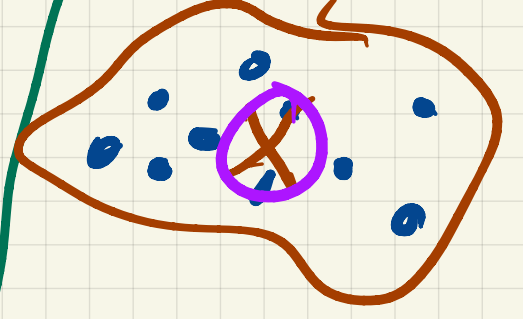
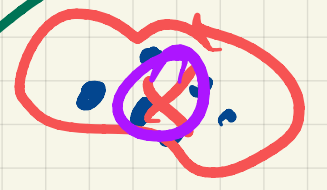
The list of embeddings is the list of keys.

Keys

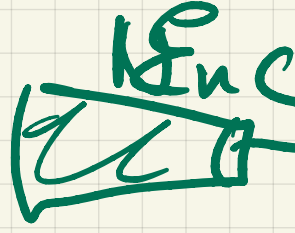
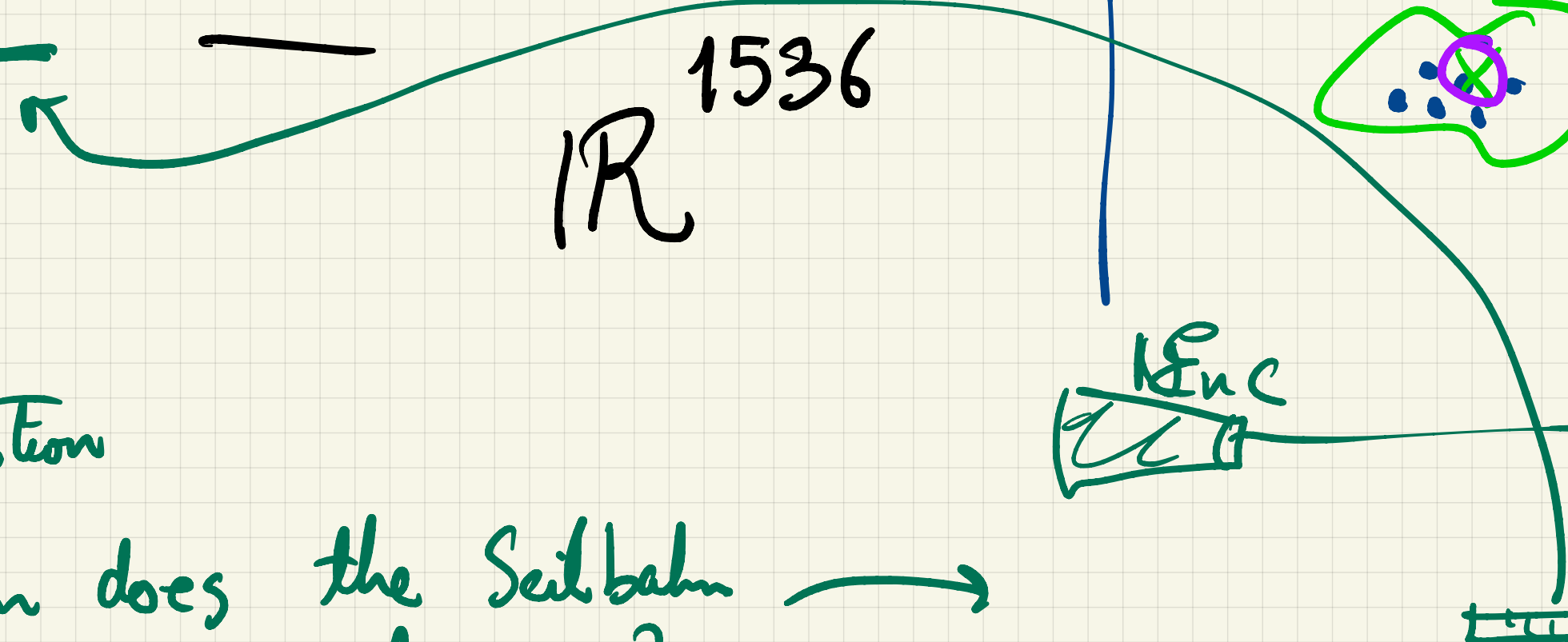


R^2

K-Means

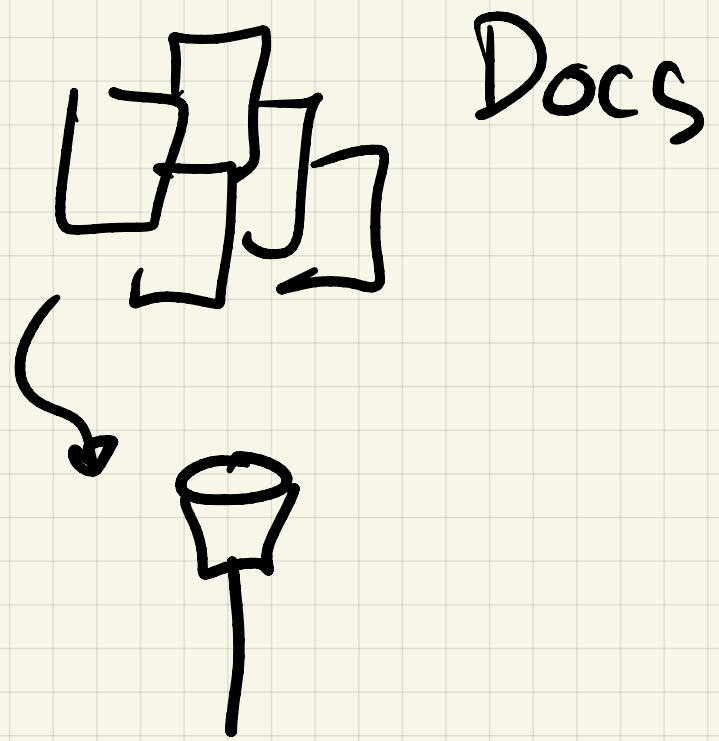


IR 1536

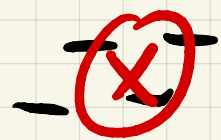
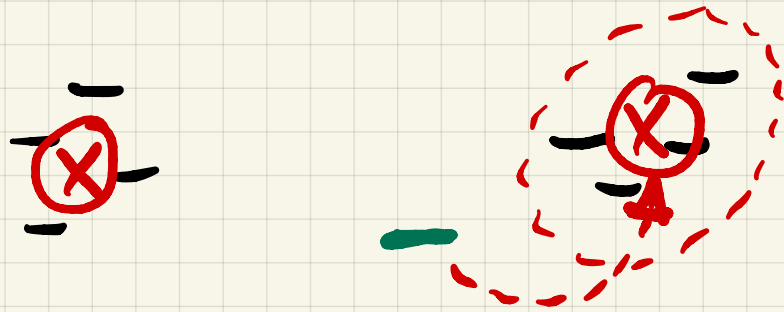


Question

When does the Seilbahn close?



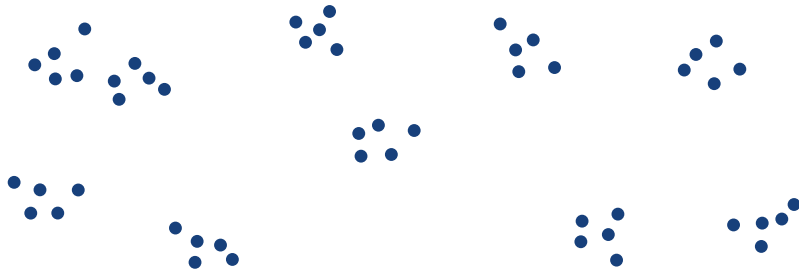
IR¹⁵³⁶



Query  

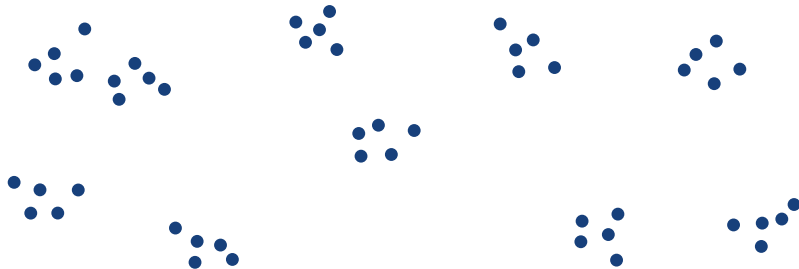
The word "Query" is written in green. To its right is a green icon consisting of a square with a circle inside, followed by a green horizontal line.

How do we effectively find the relevant keys?



ifty points

How do we effectively find the relevant keys?

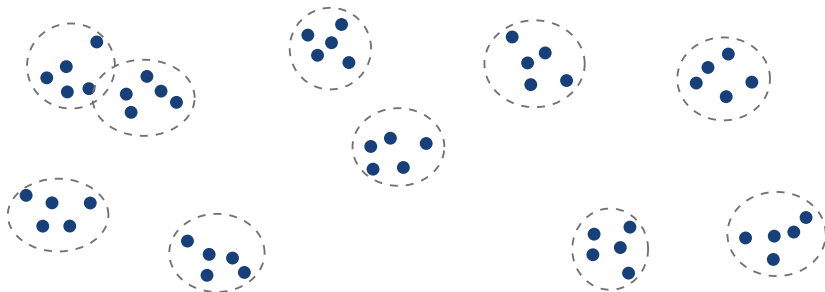


50 points

illustrative, not measured

A flattened shadow. The clumps are found in all 1536 dimensions, not in this picture.

How do we effectively find the relevant keys?

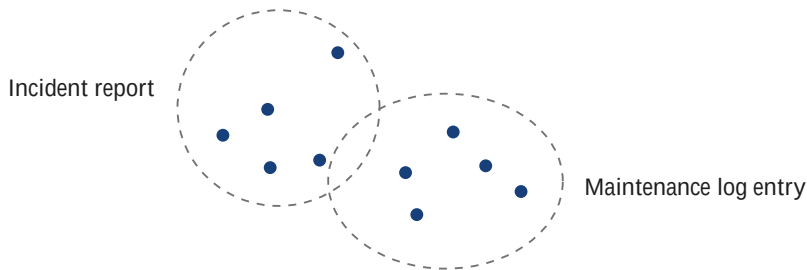


ifty points, ten clumps

illustrative, not measured

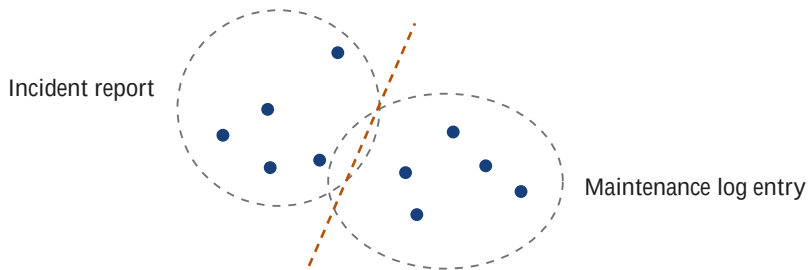
A flattened shadow. The clumps are found in all 1536 dimensions, not in this picture.

Two clumps that touch



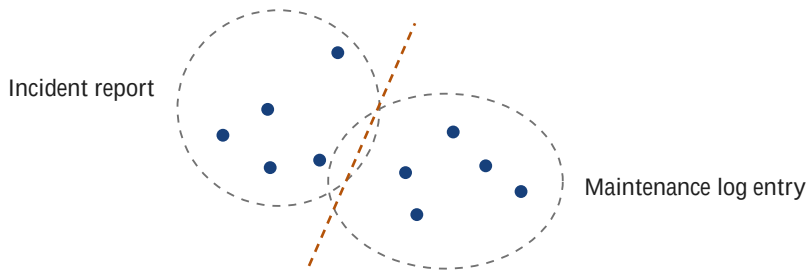
illustrative

Two clumps that touch



illustrative

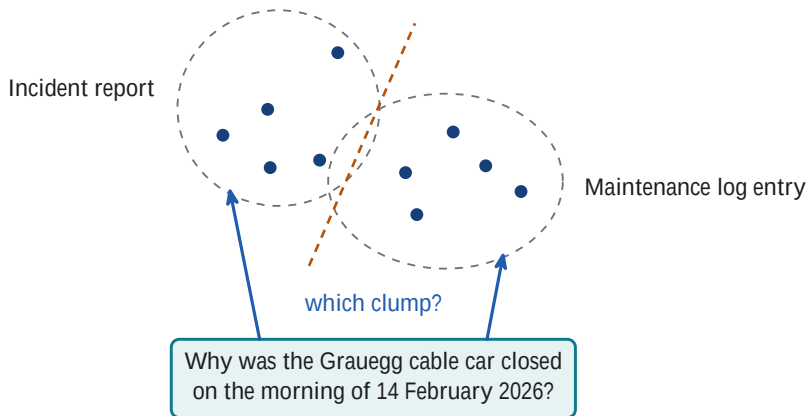
Two clumps that touch



illustrative

Why was the Grauegg cable car closed on the morning of 14 February 2026?

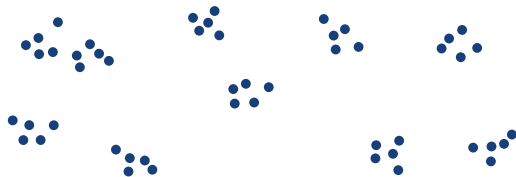
Two clumps that touch



illustrative

Finding the relevant keys

Done once, before any question

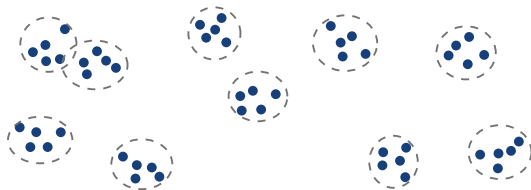


□fty embeddings

illustrative

Done once, before any question

(k-means, the one most of you have run)

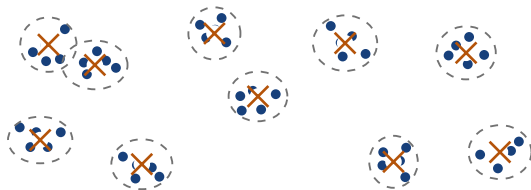


□ 50 embeddings

illustrative

Done once, before any question

(k-means, the one most of you have run)



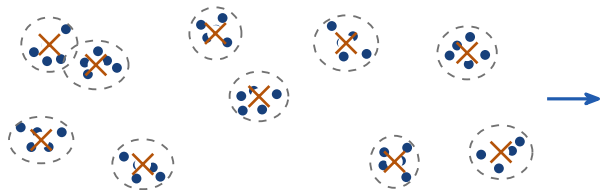
□ 50 embeddings

illustrative

Finding the relevant keys

Done once, before any question

(k-means, the one most of you have run)



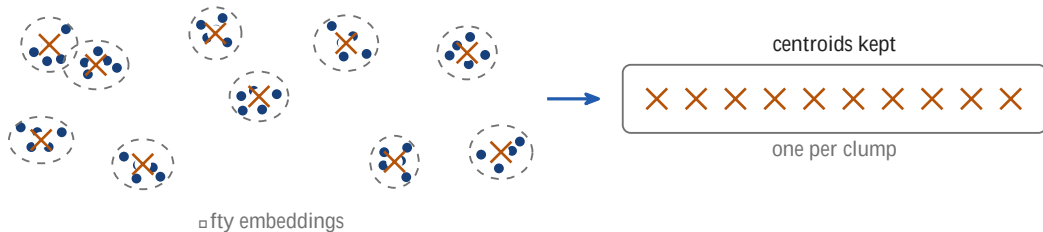
centroids kept

□fty embeddings

illustrative

Done once, before any question

(k-means, the one most of you have run)



illustrative

the pause between rounds, when nobody was asking you anything

At question time

How much do we get back for each extra day
a consignment is overdue?



E

At question time

How much do we get back for each extra day
a consignment is overdue?



E



the query lands
in the same space



At question time

How much do we get back for each extra day
a consignment is overdue?



E



the query lands
in the same space



ten comparisons

At question time

How much do we get back for each extra day
a consignment is overdue?

E

the query lands
in the same space



ten comparisons

Matcher

the question $\mathbf{x} = [0.21 \quad -0.04 \quad \dots]$

At question time

How much do we get back for each extra day
a consignment is overdue?

E

the query lands
in the same space



ten comparisons

Matcher

the question $\mathbf{x} = [0.21 \quad -0.04 \quad \dots]$

one candidate centroid $\mathbf{y} = [0.19 \quad -0.07 \quad \dots]$

At question time

How much do we get back for each extra day
a consignment is overdue?

E

the query lands
in the same space



ten comparisons

Matcher

the question $\mathbf{x} = [0.21 \quad -0.04 \quad \dots]$

one candidate centroid $\mathbf{y} = [0.19 \quad -0.07 \quad \dots]$

$$\mathbf{x}^T \mathbf{y} = \sum_i x_i y_i$$

At question time

How much do we get back for each extra day
a consignment is overdue?

E

the query lands
in the same space



ten comparisons

Matcher

the question $\mathbf{x} = [0.21 \quad -0.04 \quad \dots]$

one candidate centroid $\mathbf{y} = [0.19 \quad -0.07 \quad \dots]$

$$\mathbf{x}^T \mathbf{y} = \sum_i x_i y_i \rightarrow$$

0.86

At question time

How much do we get back for each extra day
a consignment is overdue?

E

the query lands
in the same space



ten comparisons

Matcher

the question $\mathbf{x} = [0.21 \quad -0.04 \quad \dots]$

one candidate centroid $\mathbf{y} = [0.19 \quad -0.07 \quad \dots]$

$$\mathbf{x}^T \mathbf{y} = \sum_i x_i y_i \rightarrow$$

illustrative

0.86

the machine's version
of partial credit

At question time

How much do we get back for each extra day
a consignment is overdue?

E

the query lands
in the same space



ten comparisons

one clump opened

Matcher

the question $\mathbf{x} = [0.21 \quad -0.04 \quad \dots]$

one candidate centroid $\mathbf{y} = [0.19 \quad -0.07 \quad \dots]$

$$\mathbf{x}^T \mathbf{y} = \sum_i x_i y_i \rightarrow$$

illustrative

0.86

the machine's version
of partial credit

At question time

How much do we get back for each extra day
a consignment is overdue?

E

the query lands
in the same space



ten comparisons



one clump opened

Matcher

the question $\mathbf{x} = [0.21 \quad -0.04 \quad \dots]$

one candidate centroid $\mathbf{y} = [0.19 \quad -0.07 \quad \dots]$

$$\mathbf{x}^T \mathbf{y} = \sum_i x_i y_i \rightarrow$$

illustrative

0.86

the machine's version
of partial credit

At question time

How much do we get back for each extra day
a consignment is overdue?

E

the query lands
in the same space



ten comparisons

one clump opened



a few comparisons

Matcher

the question $\mathbf{x} = [0.21 \quad -0.04 \quad \dots]$

one candidate centroid $\mathbf{y} = [0.19 \quad -0.07 \quad \dots]$

$$\mathbf{x}^T \mathbf{y} = \sum_i x_i y_i \rightarrow$$

illustrative

0.86

the machine's version
of partial credit

At question time

How much do we get back for each extra day
a consignment is overdue?

E

the query lands
in the same space



ten comparisons



one clump opened

five comparisons

Matcher

the question $\mathbf{x} = [0.21 \quad -0.04 \quad \dots]$

one candidate centroid $\mathbf{y} = [0.19 \quad -0.07 \quad \dots]$

$$\mathbf{x}^T \mathbf{y} = \sum_i x_i y_i \rightarrow$$

illustrative

0.86

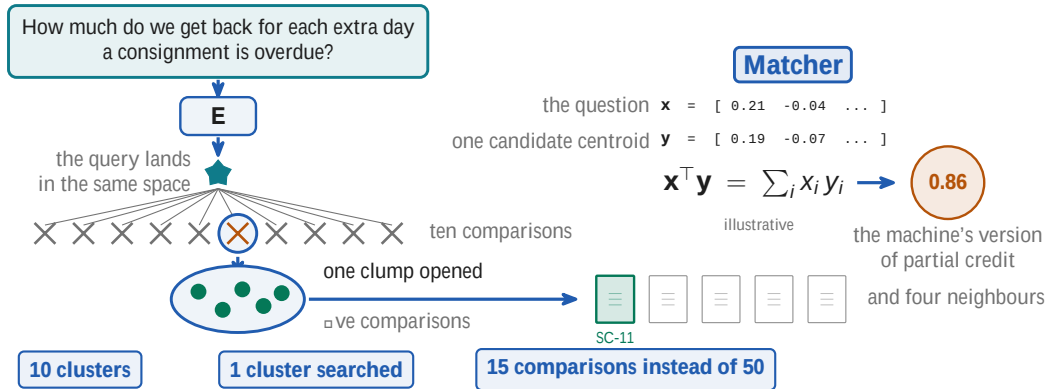
the machine's version
of partial credit
and four neighbours



SC-11



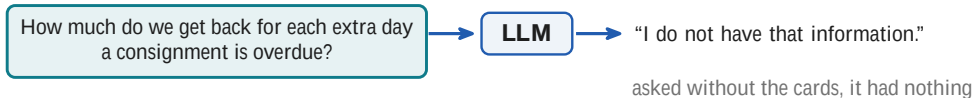
At question time



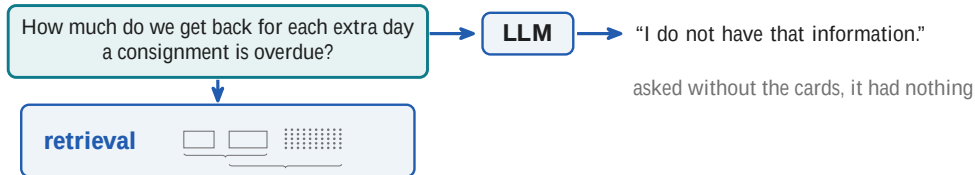
The full system

How much do we get back for each extra day
a consignment is overdue?

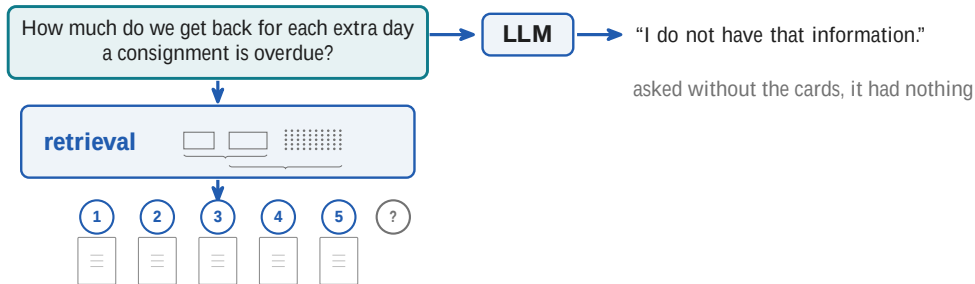
The full system



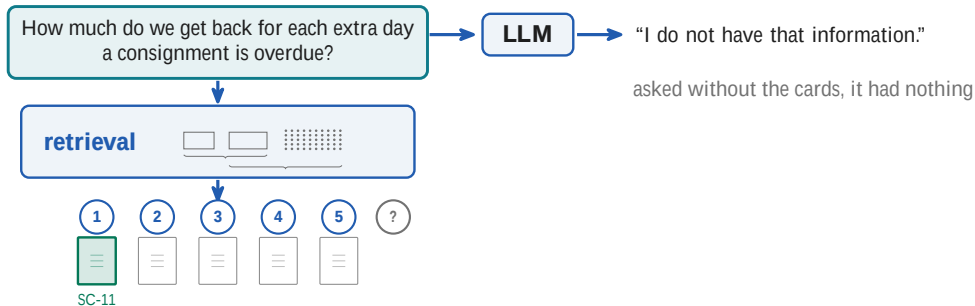
The full system



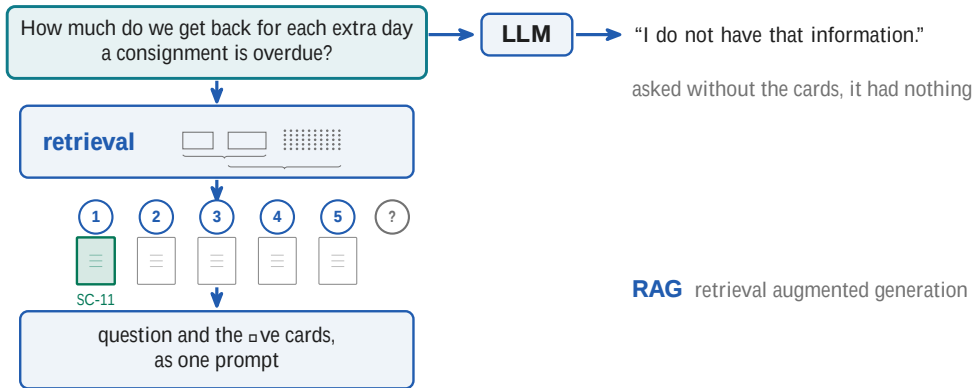
The full system



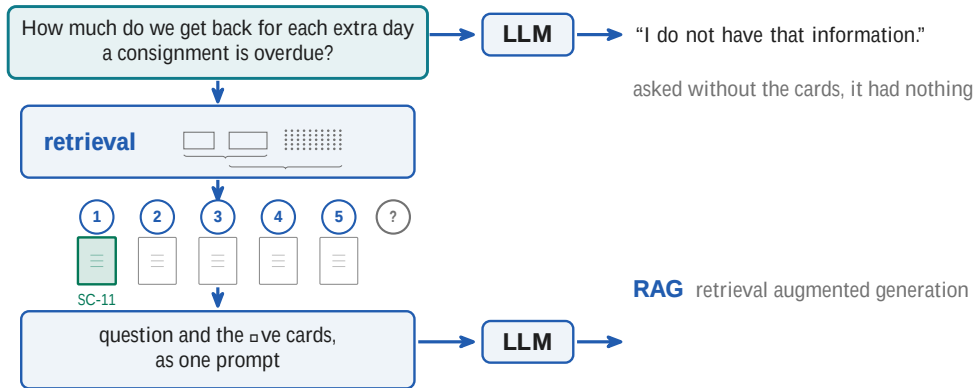
The full system



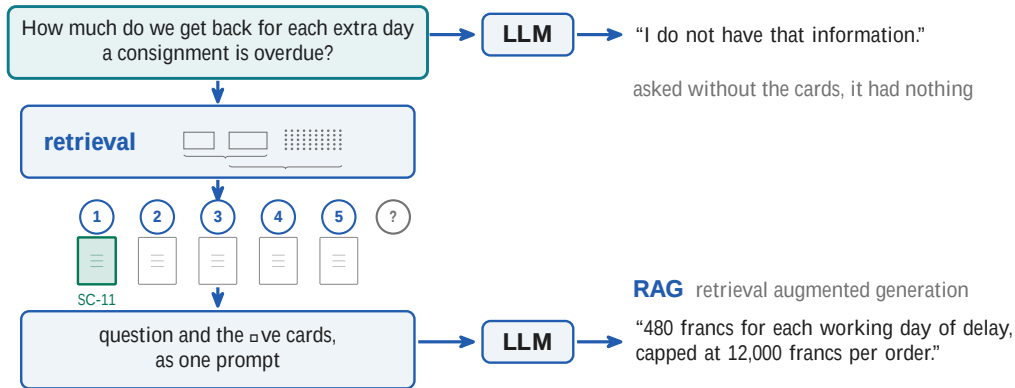
The full system



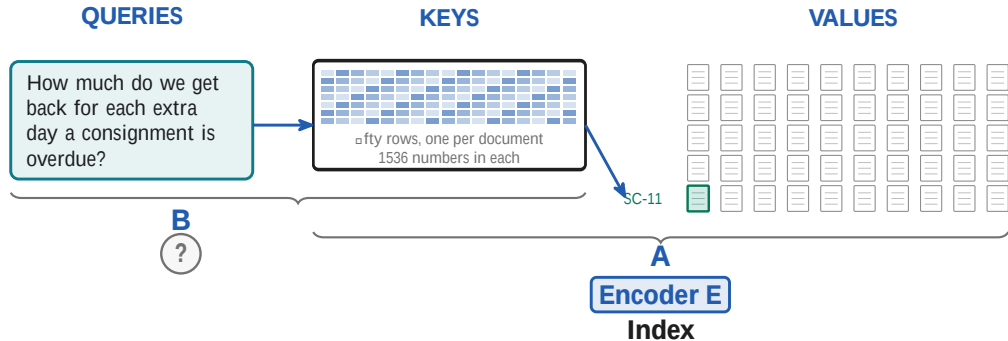
The full system



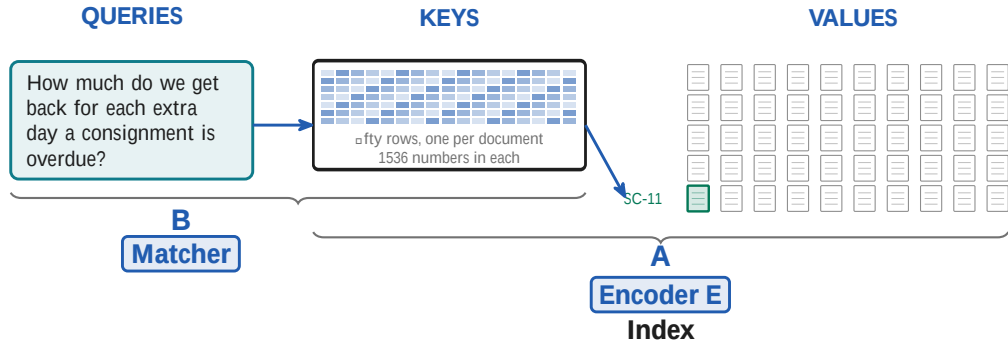
The full system



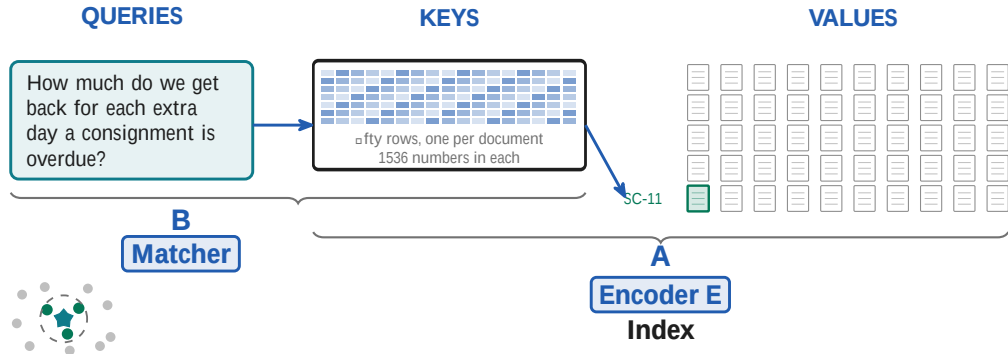
Queries, keys and values



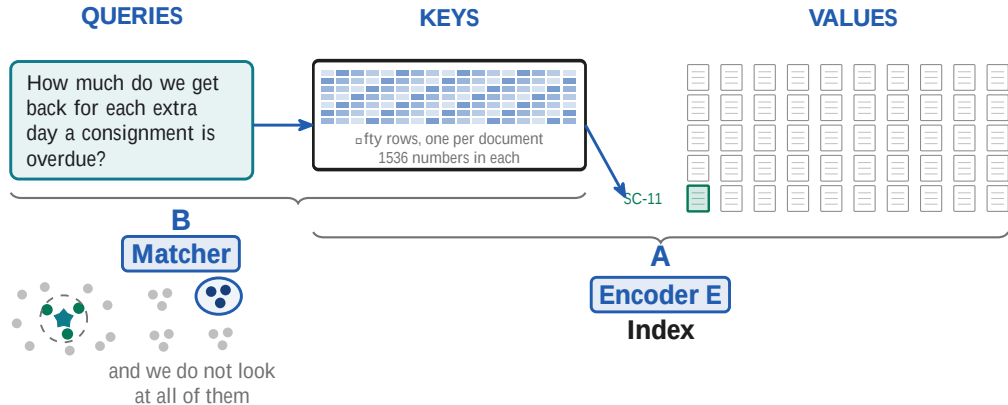
Queries, keys and values



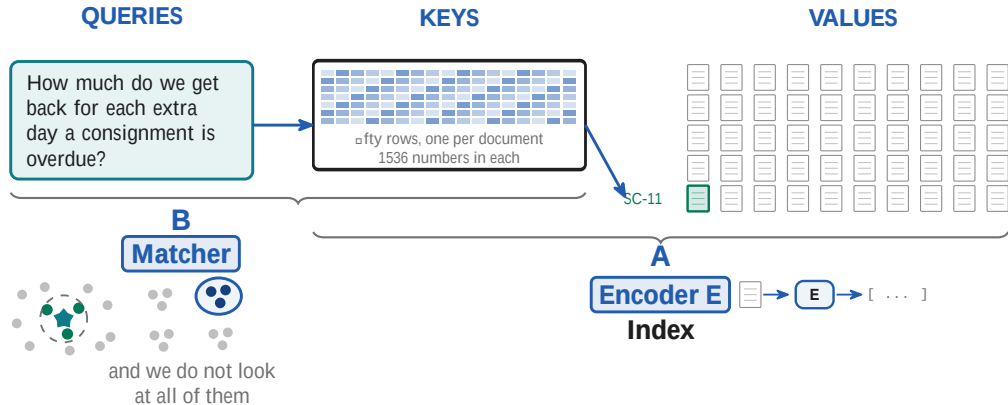
Queries, keys and values



Queries, keys and values



Queries, keys and values



Queries, keys and values

